NAME: _____ Geometry (FL B.E.S.T.)

8.4 Perimeter and Area on the Coordinate Plane

On the coordinate plane, the distance formula gives every side length, so perimeter is just a sum of distances. For area, axis-aligned rectangles and right triangles can be read directly off the grid, and figures with horizontal bases can use the familiar area formulas. For an irregular polygon whose vertices are not conveniently aligned with the axes, use the **Shoelace Formula**: list the vertices in order around the polygon, then $A = \frac{1}{2}\,|\sum(x_i y_{i+1} - x_{i+1} y_i)|$ — multiply each x by the next y, subtract the next x times the current y, add up all these terms, take the absolute value, and halve it. This worksheet builds from basic axis-aligned figures up to the Shoelace Formula for general polygons.

Practice Set — All Questions

15 questions. Work them on your own paper, then check the key below.

1. Rectangle $(0, 0)$, $(5, 0)$, $(5, 3)$, $(0, 3)$. Perimeter?

2. Same rectangle. Area?

3. $(0, 0)$, $(10, 0)$, $(0, 4)$. Area?

4. $(0, 0)$, $(6, 0)$, $(0, 8)$. Perimeter?

5. Side endpoints $(1, 1)$ and $(1, 6)$. Area of the square?

6. $(2, 3)$ to $(8, 11)$. Length?

7. Rectangle width 7, height 2. Area?

8. Square side 9. Perimeter?

9. $(0, 0)$, $(12, 0)$, $(0, 5)$. Area?

10. Bases 6 and 10, height 4. Area?

11. Rhombus side 7. Perimeter?

12. $(1, 1)$, $(7, 1)$, $(7, 4)$, $(1, 4)$. Area?

NAME: _____ Geometry (FL B.E.S.T.)

13. $(0, 0)$, $(5, 0)$, $(5, 12)$. Perimeter of the triangle?

14. Use the Shoelace Formula for vertices (in order) $(0, 0)$, $(6, 1)$, $(5, 5)$, $(1, 4)$.

15. Vertices $(2, 3)$, $(7, 3)$, $(5, 9)$. Find the area two ways: using base × height, and using the Shoelace Formula. Confirm they match.

✓ Answer Key

1. 16
2. 15
3. 20
4. $6 + 8 + 10 = 24$
5. 25
6. 10
7. 14
8. 36
9. 30
10. 32
11. 28
12. 18
13. $5 + 12 + 13 = 30$
14. 20 square units
15. Both methods give 15 square units